1. Assume that the markets for sugar cane, rum, and whiskey are initially in equilibrium. Assume further that Hurricane Marilyn destroys much of the Jamaican sugar cane crop. Sugar cane is a principal ingredient in rum, but it is not an ingredient in whiskey. Analyze the effect of the hurricane on the markets for each of the three goods. Explain using graphs.

2. Consider the market for hotdogs (normal good) and butter bun (inferior good). If the government imposes a 25% increase in income tax (causing after-tax income to fall), what is expected to happen to the equilibrium price and quantity for hotdogs and butter bun?

3. Consider the following information for the demand and supply

   $Q_s = 10 + 3P$
   $Q_d = 40 - 2P$

   where Q_s: Quantity supplied & Q_d: Quantity demanded

   a) Find the market equilibrium price (P) at which there will be no excess or surplus in the market.
   b) If government decides to fix the price at P = 4, calculate the shortage/surplus.

4. Suppose that the market demand for potatoes is given by $Q = 1000 - 250P$ and the market supply of potatoes is given by $Q = 150P$, where P is the price per bag of potatoes and Q is the number of bags per month.

   a) What is the equilibrium price and quantity for potatoes? Illustrate your answer graphically.
   b) Recently, a popular cooking show featured several appealing and easy-to-make potato dishes, leading to a surge in consumer interest in potatoes. Assuming that the market is initially at equilibrium, explain how this scenario would affect the market for potatoes using a demand and supply diagram.
   c) Now consider that several potato-producing regions experience a severe outbreak of crop-destroying pests, which significantly impacts the overall potato harvest. In addition, the cost of pesticides and pest control measures rises sharply due to high demand. Assuming that the market is initially at equilibrium explain how this scenario would affect the market for potatoes, using a demand and supply diagram to illustrate your answer.
   d) Given the scenarios in part (b) and (c), if the effect of pest infestation outweighs effect of surge in consumer interest in potatoes, explain how the market equilibrium price and quantity for potatoes will be affected using demand and supply diagram.